

# Mina Tahmasbi Arashloo

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## Research Interest

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Networked systems, with an emphasis on software-defined and programmable networks.

## Education

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| <b>Princeton University</b><br>Ph.D. and M.A in Computer Science<br>Advisor: Jennifer Rexford<br>Thesis: Stateful Programming of High-Speed Network Hardware | 2014 - 2019 |
| <b>Sharif University of Technology</b><br>B.Sc. in Computer Engineering (GPA: 19.50/20.00)                                                                   | 2010 - 2014 |

## Employment

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| <b>University of Waterloo</b><br>Cheriton School of Computer Science<br>Assistant Professor                                                      | 2022 - present |
| <b>Cornell University</b><br>Department of Computer Science<br>Presidential Post-Doctoral Fellow<br>Supervised by Nate Foster and Rachit Agarwal | 2019 - 2022    |
| <b>Microsoft Research</b><br>Research Intern                                                                                                     | Summer 2017    |
| <b>Microsoft Azure</b><br>Intern                                                                                                                 | Fall 2016      |

## Honors and Recognitions

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|-----------------------------------------------------------------------------------------|----------------|
| Canada Research Chair in “Minimizing Human Error in Operating Modern Networks” (Tier 2) | 2024 - present |
| Heidelberg Laureate Forum Young Researcher                                              | 2023           |
| NSERC Discovery Grant Early Career Supplement                                           | 2023           |
| Rising Star in Networking and Communications by N <sup>2</sup> Women                    | 2021           |
| Best Paper Award at ACM SIGCOMM SOSR                                                    | 2020           |
| Cornell Presidential Post-Doctoral Fellow                                               | 2019 - 2022    |
| ACM SIGCOMM Doctoral Dissertation Award                                                 | 2019           |
| Siebel Scholar                                                                          | Class of 2019  |
| Microsoft Research Dissertation Grant                                                   | 2019           |
| EECS Rising Stars in EECS at MIT                                                        | 2018           |
| School of Engineering and Applied Science Award of Excellence, Princeton University     | 2017           |

## Publications

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### *A Protocol-Independent Transport Architecture*

Kimiya Mohammadtaheri, David Gao, Samuel Zhang, Matthew Chen, Eric Su, Pengyu Ji, Saad Syed, Chris Neely, Mario Baldi, Nachiket Kapre, Mina Tahmasbi Arashloo  
Pre-print, Under Submission

### *High-Level and Target-Agnostic Transport Programs*

Pedro Mizuno, Kimiya Mohammadtaheri, Linfan Qian, Joshua Johnson, Danny Akbarzadeh, Chris Neely, Mario Baldi, Nachiket Kapre, Mina Tahmasbi Arashloo  
Pre-print, Under Submission

### *Count-Based Abstractions for Performance Verification of Contention Points*

Amir Seyhani, Aarti Gupta, David Walker, Mina Tahmasbi Arashloo  
USENIX Symposium on Networked Systems Design and Implementation, **NSDI 2026**

### *Analyzing Symbolic Properties for DRL Agents in Systems and Networking*

Mohammad Zangoeei, Jannis Weil, Amr Rizk, Mina Tahmasbi Arashloo, Raouf Boutaba  
ACM Special Interest Group on Performance Evaluation, **ACM SIGMETRICS 2026**

### *Dynamic SLA-Aware Network Slice Monitoring*

Niloy Saha, Mina Tahmasbi Arashloo, Nashid Shahriar, Raouf Boutaba  
ACM Special Interest Group on Performance Evaluation, **ACM SIGMETRICS 2026**

### *Toward eBPF-Accelerated Pub-Sub Systems*

Beihao Zhou, Samer Al-Kiswany, Mina Tahmasbi Arashloo  
ACM SIGCOMM eBPF Workshop, **eBPF 2025**

### *Buffy: A Formal Language-Based Framework for Network Performance Analysis*

Amir Seyhani, Junyi Zhao, Aarti Gupta, David Walker, Mina Tahmasbi Arashloo  
ACM SIGCOMM Workshop on Hot Topics in Networks, **HotNets 2024**

### *Formal Methods for Network Performance Analysis*

Mina Tahmasbi Arashloo, Ryan Beckett, Rachit Agarwal  
USENIX Symposium on Networked Systems Design and Implementation, **NSDI 2023**

### *dcPIM: Near-Optimal Proactive Datacenter Transport*

Qizhe Cai, Mina Tahmasbi Arashloo, Rachit Agarwal  
ACM Special Interest Group on Data Communication, **SIGCOMM 2022**

### *Modular Switch Programming under Resource Constraints*

Mary Hogan, Shir Landau-Feibish, Mina Tahmasbi Arashloo, Jennifer Rexford, David Walker  
USENIX Symposium on Networked Systems Design and Implementation, **NSDI 2022**

### *Towards Formally Verifying Congestion Control Behavior*

Venkat Arun, Mina Tahmasbi Arashloo, Ahmed Saeed, Mohammad Alizadeh, Hari Balakrishnan  
ACM Special Interest Group on Data Communication, **SIGCOMM 2021**

### *DBVal: Validating P4 Data Plane Runtime Behavior*

K Shiv Kumar, Ranjitha K, P S Prashanth, Mina Tahmasbi Arashloo, Venkanna U., Praveen Tammana  
ACM SIGCOMM Symposium on SDN Research, **SOSR 2021**

### *Petr4: Formal Foundations for P4 Data Planes*

Ryan Doenges, Mina Tahmasbi Arashloo, Santiago Bautista, Alexander Chang, Newton Ni, Samwise Parkinson, Rudy Peterson, Alaia Solko-Breslin, Amanda Xu, Nate Foster  
ACM SIGPLAN Symposium on Principles of Programming Languages, **POPL 2021**

### *Elastic Switch Programming with P4All*

Mary Hogan, Shir Landau-Feibish, Mina Tahmasbi Arashloo, Jennifer Rexford, David Walker, Rob Harrison  
ACM Workshop on Hot Topics in Networks, **HotNets 2020**

### *Enabling Programmable Transport Protocols in High-Speed NICs*

Mina Tahmasbi Arashloo, Alexey Lavrov, Manya Ghobadi, Jennifer Rexford, David Walker, David Wentzlaff

USENIX Symposium on Networked Systems Design and Implementation, **NSDI 2020**

*Tracking P4 Program Execution Path in the Data Plane*

Suriya Kodeswaran, Mina Tahmasbi Arashloo, Praveen Tammana, Jennifer Rexford

ACM SIGCOMM Symposium on SDN Research, **SOSR 2020 (Best Paper Award)**

*A Scalable VPN Gateway for Multi-Tenant Cloud Services*

Mina Tahmasbi Arashloo, Pavel Shirshov, Rohan Gandhi, Guohan Lu, Lihua Yuan, Jennifer Rexford

ACM SIGCOMM Computer Communication Review, **SIGCOMM CCR 2018**

*HotCocoa: Hardware Congestion Control Abstractions*

Mina Tahmasbi Arashloo, Monia Ghobadi, Jennifer Rexford, David Walker

ACM Workshop on Hot Topics in Networks, **HotNets 2017**

*SNAP: Stateful Network-wide Abstractions for Packet Processing*

Mina Tahmasbi Arashloo, Yaron Koral, Michael Greenberg, Jennifer Rexford, David Walker

ACM Special Interest Group on Data Communication, **SIGCOMM 2016**

*Compiling Path Queries*

Srinivas Narayana, Mina Tahmasbi Arashloo, Jennifer Rexford, David Walker

USENIX Symposium on Networked Systems Design and Implementation, **NSDI 2016**

## Teaching

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### University of Waterloo

|                                     |             |
|-------------------------------------|-------------|
| CS 856: Network Transport Protocols | Winter 2026 |
| CS 456: Computer Networks           | Fall 2025   |
| CS 856: Programmable Networks       | Winter 2025 |
| CS 456: Computer Networks           | Fall 2024   |
| CS 856: Programmable Networks       | Winter 2024 |
| CS 456: Computer Networks           | Fall 2023   |
| CS 856: Programmable Networks       | Winter 2023 |

### Cornell University

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| Online certificate series on Software-Defined Networking (SDN)                     | 2020 - 2022 |
| In collaboration with Nate Foster and Cornell Online Education Programs (eCornell) |             |

## Grants

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|---------------------------------------------------------------------------|-------------|
| <b>Mitacs Accelerate</b>                                                  | 2025        |
| Sole PI, \$39,800                                                         |             |
| <b>Canada Research Chair (Tier 2)</b>                                     | 2024 - 2029 |
| Sole PI, \$500,000 in total including \$70,000 annually in research funds |             |
| <b>NSERC Discovery Grant</b>                                              | 2023 - 2028 |
| Sole PI, \$31,000 annually                                                |             |
| <b>NSERC Discovery Grant Early Career Supplement</b>                      | 2023        |
| Sole PI, \$12,500                                                         |             |

## Professional Service

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### Program Co-Chair

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|----------------------------------------------|------|
| P4 Workshop                                  | 2023 |
| ACM SIGCOMM Symposium on SDN Research (SOSR) | 2022 |
| The N <sup>2</sup> Women Workshop at SIGCOMM | 2022 |

### Organizing Committee

|                                        |      |
|----------------------------------------|------|
| ACM SIGCOMM Non-Paper Session Co-Chair | 2026 |
|----------------------------------------|------|

|                                     |              |
|-------------------------------------|--------------|
| N2Women Workshop Steering Committee | 2025-present |
| IEEE CNSM Keynote Co-Chair          | 2023         |
| ACM SIGCOMM Publication co-chair    | 2020         |

#### **Program Committee Member**

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|-------------------------------------------------------------------------------------|------------------|
| ACM Special Interest Group on Data Communication (SIGCOMM)                          | 2026             |
| USENIX Symposium on Networked Systems Design and Implementation (NSDI)              | 2026             |
| ACM Special Interest Group on Data Communication (SIGCOMM)                          | 2025             |
| ACM Special Interest Group on Data Communication (SIGCOMM)                          | 2024             |
| USENIX Symposium on Networked Systems Design and Implementation (NSDI)              | 2024             |
| ACM Conference on emerging Networking EXperiments and Technologies (CoNEXT)         | 2024             |
| ACM Special Interest Group on Performance Evaluation (SIGMETRICS)                   | 2024             |
| ACM SIGCOMM Workshop on Hot Topics in Networks (HotNets)                            | 2024             |
| USENIX Symposium on Networked Systems Design and Implementation (NSDI)              | 2023             |
| ACM Symposium on Cloud Computing (SoCC)                                             | 2022             |
| ACM Conference on emerging Networking EXperiments and Technologies (CoNEXT)         | 2021             |
| ACM SIGCOMM Workshop on Hot Topics in Networks (HotNets)                            | 2021             |
| ACM/IEEE Symposium on Architectures for Networking and Communication Systems (ANCS) | 2021             |
| ACM Special Interest Group on Data Communication (SIGCOMM)                          | 2020             |
| ACM SIGCOMM Symposium on SDN Research (SOSR)                                        | 2019 - 2022      |
| ACM SIGCOMM Posters and Demos                                                       | 2020             |
| Asia-Pacific Workshop on Networking (APNet)                                         | 2020, 2022       |
| EuroP4 Workshop                                                                     | 2020, 2021       |
| P4 Workshop                                                                         | 2018, 2019, 2021 |

#### **Other Committees and Panels**

|                                        |            |
|----------------------------------------|------------|
| IRTF Applied Networking Research Prize | 2025, 2026 |
| NSF CNS Panel                          | 2024       |
| NSF CISE Panel                         | 2023       |
| NSF CNS Panel                          | 2020       |

#### **External Reviewer for Conference**

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|----------------------------------------------------------|------|
| ACM SIGPLAN ASPLOS (External Review Committee)           | 2021 |
| IEEE International Conference on Computer Communications | 2017 |

#### **External Reviewer for Journal**

|                                                                   |             |
|-------------------------------------------------------------------|-------------|
| IEEE/ACM Transactions on Networking (TON)                         | 2018 - 2022 |
| IEEE Transactions on Network and Service Management (TNSM)        | 2021        |
| IEEE Transactions on Very Large Scale Integration Systems (TVLSI) | 2018        |
| Journal of Cloud Computing                                        | 2017        |

#### **University Service**

|                                            |                       |
|--------------------------------------------|-----------------------|
| Women in Computer Science (WiCS) committee | August 2022 - present |
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### **Student Supervision and Mentoring**

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#### **Graduate Students at University of Waterloo**

Nesa Abbasimoghadamniassar, MMath  
 Amir Hossein Seyhani, Ph.D.  
 Kimiya MohammadTaheri, MMath → Ph.D  
 Pedro Mizuno, MMath → Ph.D.

#### **Undergraduate Students**

### *Full Time*

|                                                        |                        |
|--------------------------------------------------------|------------------------|
| Andrew Qiu, Undergraduate Researcher                   | Spring 2026            |
| Pradhyumn Varadhan, Undergraduate Research Fellow, UBC | Fall 2024, Spring 2025 |
| Linfan Qian, Undergraduate Research Fellow             | Fall 2024              |
| Jamie Kai, Undergraduate Research Fellow               | Fall 2024              |
| Luc Edes, Undergraduate Research Fellow                | Winter 2024            |

### *Part Time*

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|-----------------------------------------------------------------------|--------------------------|
| Andrew Qiu, Undergraduate Research Assistant                          | Winter 2026              |
| Matthew Chen, Undergraduate Research Assistant                        | Winter 2026              |
| Alex Xue, Undergraduate Research Assistant                            | Fall 2025                |
| Beihao Zhou, Undergraduate Research Assistant (with Samer Al-Kiswany) | Winter 2024, Spring 2025 |
| Joshua Johnson, Undergraduate Research Assistant                      | Winter 2025              |
| Linfan Qian, Undergraduate Research Assistant                         | Winter 2025              |
| Zening Hao, Undergraduate Research Assistant                          | Winter 2025              |
| Joshua Johnson, Undergraduate Research Assistant                      | Spring 2024              |
| Alex Lin, Undergraduate Research Assistant                            | Spring 2024              |
| Patrick He, Undergraduate Research Assistant                          | Spring 2024              |
| Danny Akbarzadeh, Undergraduate Research Assistant                    | Spring 2024              |
| Carol Duan, Undergraduate Research Assistant                          | Winter 2024              |
| Crystal Zhou, Undergraduate Research Assistant                        | Winter 2024              |
| Zening Hao, Undergraduate Research Assistant                          | Winter 2024              |
| Arnav Nagpal, Undergraduate Research Assistant                        | Winter 2024              |
| Jiashuo Shi, Undergraduate Research Assistant                         | Winter 2024              |
| Harshvardhan Kakar, Undergraduate Research Assistant                  | Spring 2023              |
| Andrew Zang, Undergraduate Research Assistant                         | Spring 2023              |
| Harshvardhan Kakar, Undergraduate Research Assistant                  | Spring 2023              |
| Dao Bo Yang, Undergraduate Research Assistant                         | Winter 2023              |
| Ruiling Ma, Undergraduate Research Assistant                          | Winter 2023              |
| Kevin Bian, Undergraduate Research Assistant                          | Winter 2023              |
| Eric Liu, Undergraduate Research Assistant                            | Fall 2022                |

### **Mentorship at Other Institutions**

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|--------------------------------------------------------------|-----------|
| Qizhe Cai, Ph.D. student at Cornell University               | 2019-2022 |
| Suriya Kudeswaran, M.Sc. student at Princeton University     | 2018-2019 |
| Melissa Ginaldi, Undergraduate student at Cornell University | 2019-2020 |

### **Outreach**

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|---------------------------------------------------------------------------------------------|------|
| Panelist, "Thesis Writing and Job/Scholarship Applications", UWaterloo's Women in Math      | 2025 |
| Panelist, "International STEM Faculty Careers", Princeton University's GradFUTURES          | 2025 |
| Panelist, "Student & Early Career Networking Researcher Roundtable", the Networking Channel | 2023 |
| Co-Organizer of the "Introduction to Network Verification" tutorial at ACM SIGCOMM          | 2021 |
| Moderator of the "Women at Microsoft" panel at ACM SIGCOMM                                  | 2021 |
| Panelist on the "Women in STEM" panel, Rutgers University - Newark Chemistry Society        | 2021 |

### **Invited Talks**

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|------------------------------------------------------------------------------------------------------------------|---------------|
| P4 Workshop, Keynote<br><i>High-Level and Target-Agnostic Transport Programs</i>                                 | October 2025  |
| University of California at Berkeley, System Seminar<br><i>High-Level and Target-Agnostic Transport Programs</i> | October 2025  |
| IEEE NetSoft, Keynote<br><i>Efficient Abstractions for Network Quality of Service</i>                            | June 2025     |
| University of Windsor, School of Computer Science Colloquium Series                                              | February 2025 |

*Towards automated proactive analysis of network performance*

|                                                                                                                                               |               |
|-----------------------------------------------------------------------------------------------------------------------------------------------|---------------|
| Dagstuhl Seminar on Network Calculus<br><i>Logic-Based Analysis of Network Performance</i>                                                    | April 2024    |
| ETH (Swiss Federal Institute of Technology) Zurich<br><i>Formal Methods for Network Performance Analysis</i>                                  | May 2023      |
| Networking and Distributed Systems Lab at HP Labs<br><i>Formal Methods for Network Performance Analysis</i>                                   | May 2023      |
| USENIX Symposium on Networked Systems Design and Implementation (NSDI)<br><i>Formal Methods for Network Performance Analysis</i>              | April 2023    |
| University of Wisconsin Internet and Systems Research Talk Series<br><i>Enabling Programmable Transport Protocols on High-Speed NICs</i>      | April 2021    |
| Rutgers CS Systems Research Seminar<br><i>Enabling Programmable Transport Protocols on High-Speed NICs</i>                                    | November 2020 |
| USENIX Symposium on Networked Systems Design and Implementation (NSDI)<br><i>Enabling Programmable Transport Protocols on High-Speed NICs</i> | February 2020 |
| Guest lecture at the Computer Networks course at MIT<br><i>An Introduction to SmartNICs and their Use Cases</i>                               | November 2019 |
| Women Advancing Research, Princeton SheRoars Conference<br><i>Flexible Network Monitoring and Control</i>                                     | October 2018  |
| ACM Workshop on Hot Topics in Networks<br><i>HotCocoa: Hardware Congestion Control Abstractions</i>                                           | November 2017 |
| University of Washington Systems Seminar<br><i>SNAP: Stateful Network-wide Abstractions for Packet Processing</i>                             | July 2017     |
| Stanford University Networking Seminar<br><i>SNAP: Stateful Network-wide Abstractions for Packet Processing</i>                               | May 2017      |
| Network Programming Initiative (NPI) Webinar Series<br><i>SNAP: Stateful Network-wide Abstractions for Packet Processing</i>                  | March 2017    |